



Smart Trading: Unlocking Market Insights with Transformer Models

Yuan Ci, Haoruo Liu, Yin Piao, Mingrui Yang, Chenrui Xu, Zeping Sheng



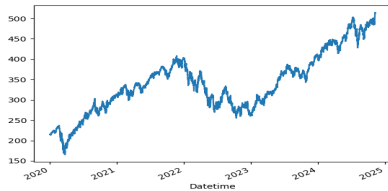
Problem Statement

Financial markets are driven by a variety of factors such as policy changes, unexpected events, and market sentiment, making it difficult for traditional econometric and machine learning models to accurately predict future trends. Therefore, we try to develop more sophisticated methods to address these issues, making contributions to optimal investing.

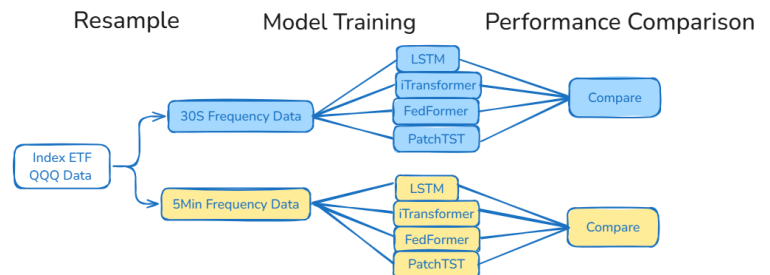
Our Impact

1. Improve forecasting accuracy and optimize investment decisions
2. Facilitate market risk management
3. Promote the development of high-frequency quantitative trading
4. Improve cross-market analytical skills

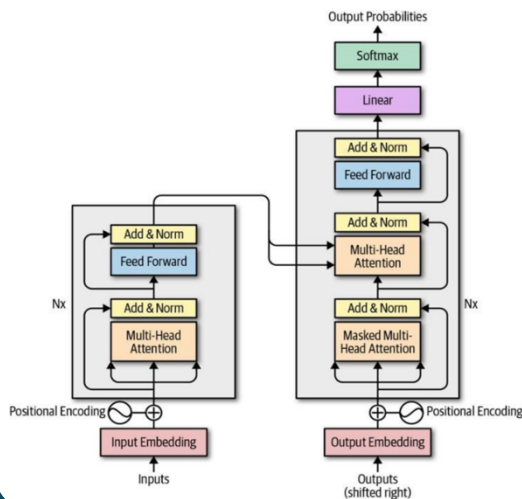
Utilize Index ETF QQQ Data



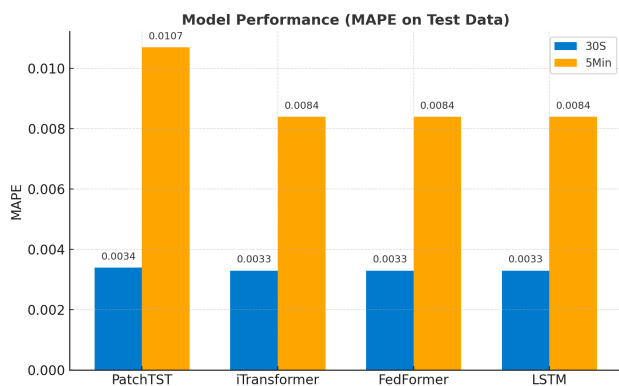
Workflow: Data → Models → Evaluation



Transformer Architecture



Models Achieve Satisfying Performance



At 30s, all models perform similarly (MAPE ~0.0033–0.0034). At 5min, PatchTST's error increases (0.0107), while others (~0.0084) maintain stability by capturing richer feature interactions.